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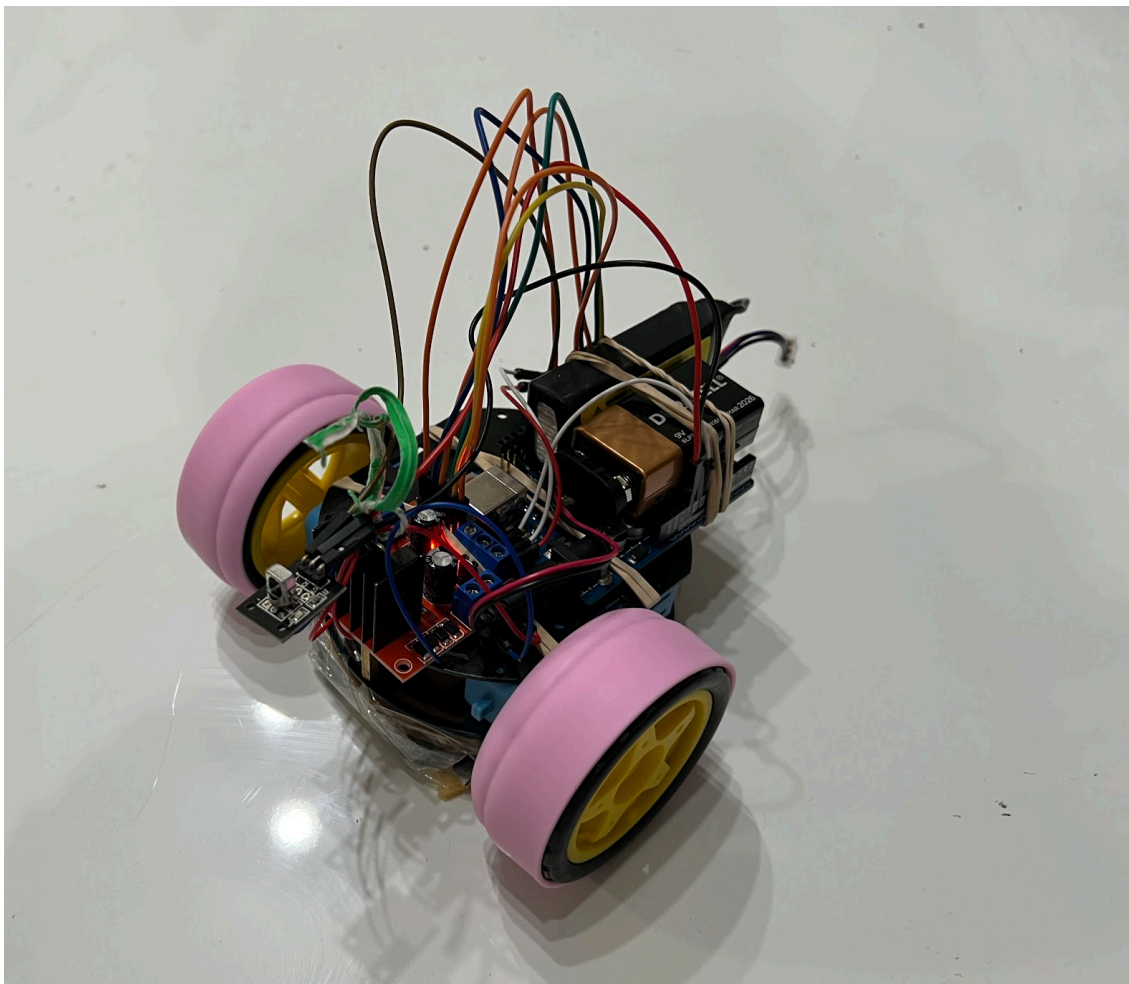
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Physics 140 Period 3

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Capstone Project Report



Video Link: <https://youtu.be/9s45LzQlnno>

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Introduction:

Have you ever felt too busy or too tired to erase the whiteboard after a long study session? Does your cheap whiteboard eraser smear your marker across the whiteboard? If so, Draggy is our solution to your problem. Draggy is an Arduino-based autonomous eraser, capable of reaching all over the whiteboard and working in the background as you prepare for your next class. **By using preset board dimensions, Draggy ensures that every nook and cranny is clean, leaving no chance for marks to cake and stain the whiteboard.** Our goal is to improve work efficiency by automating small and repetitive tasks.

As whiteboards continue to replace blackboards in modern education and business environments, a whole new class of convenience products rises – the magnetic whiteboard accessory. From marker holders to whiteboard erasers, technology is catching up to the revolution that whiteboards are. We're taking it a step further. We propose a new paradigm to disrupt the magnetic whiteboard accessory space and bring industry-changing convenience to the whiteboard user – the automatic whiteboard eraser, Draggy.

Research:

We took inspiration from Wipy, a similar project that uses time-of-flight and infrared sensors to autonomously follow whiteboard marks more efficiently. They also used long wires to connect Wipy to an external power source, something we knew we wanted to avoid for a sleeker, more convenient design.

Initial Assumptions:

Rubber RC car wheels will have suitable traction to transport Draggy across the whiteboard. The motors will be strong enough to power the wheels against the force of gravity. A 5V power bank will provide enough power to the wheels.

Design/Design Process:

We envisioned an autonomous vehicle that could glide across the whiteboard to erase marks. Our initial design included a cardboard chassis affixed to the whiteboard by magnets. Another layer of cardboard acted as a spacer between the board and the magnets to allow Draggy to move along the whiteboard. We used rubber RC car wheels powered by a 48:1 motor linked with an Arduino-based breadboard.

The cardboard chassis was too flexible, and the magnetic force would deform the edges, raising the wheels away from the whiteboard. This caused a loss of traction, and Draggy would not be able to move. To solve this, we replaced the cardboard with an acrylic chassis, which could resist deformation (See Figure 1).

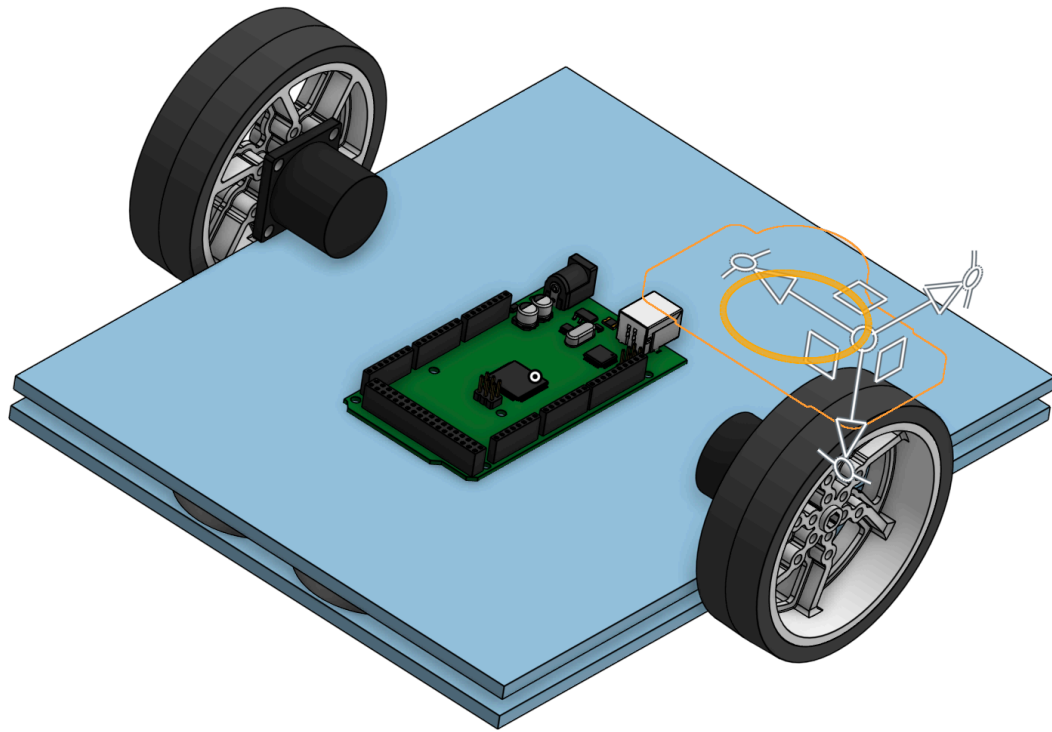


Figure 1: Initial Acrylic Design for Draggy

Even with this replacement, the wheels still lacked the power and traction to transport Draggy. To get more power, we found higher torque in a 96:1 micromotor with a metal gearbox, instead of a plastic one. The new micromotors did not have wires to connect them to the breadboard, so we stripped some wires and soldered them to the motors for convenience (See Figure 2).

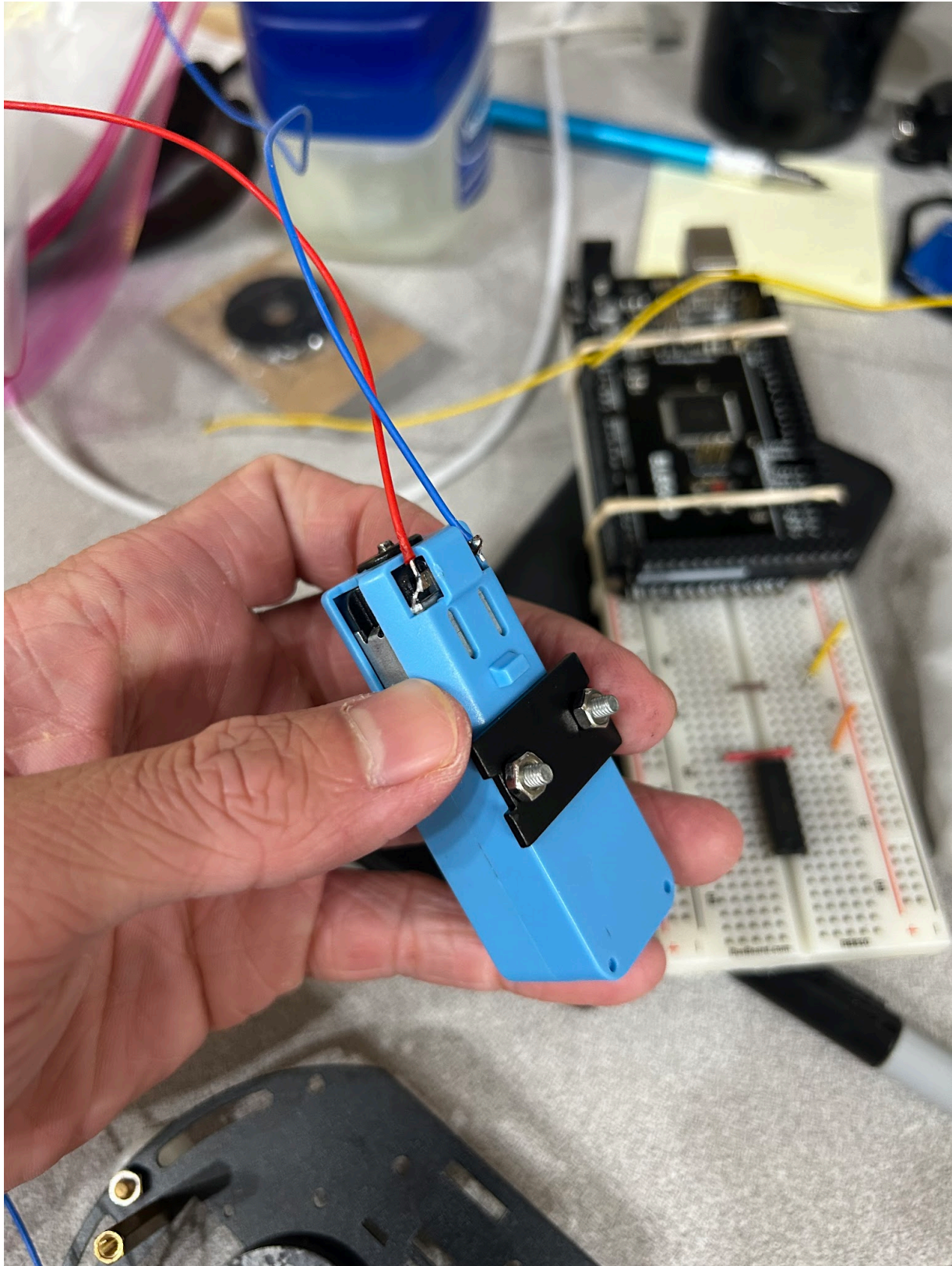


Figure 2: Soldering wires to the micromotors

Taking inspiration from Hydroflask water bottle boots, we tested a rubber-silicone coating on the wheels. Using baby bottle boots, our wheels had enough traction to transfer the motor's power into the whiteboard and move Draggy.

The motors were attached to the acrylic chassis with tape, which was unstable. We replaced the chassis with a metal bracket that could be stacked and assembled with screws. The new chassis was recycled from a middle-school robotics build. With the new chassis, the motors can be properly aligned, and Draggy has a more sturdy base.

Draggy's motors are powered with a 7.4V 1300mAh RC Car battery, and its Arduino is powered with a regular 9V battery. The product is attached to the whiteboard by magnets, which are spaced with packaging tape to stick to the whiteboard yet allow movement. Additionally, we screwed the motors to a rigid metal body, ensuring that the wheels would face straight and that Draggy would remain intact when colliding with obstacles.

We used the Arduino through an H-bridge module to control the pair of motors. After realizing that the H-bridge's current rating was far less than what we'd hoped to supply to the motors, we switched to a more robust L298D motor driver. As a failsafe mechanism to the autonomous routing around the whiteboard, we used an IR remote/receiver pair to control the motors manually. To do this, we connected the signal pin of the receiver module to the Arduino. If this pin gets a specific input, the motors are activated and vice versa.

Our final change was turning the water bottle boots inside out, which helped add traction to our wheels by increasing the surface area of the rubber.

Our final chassis design required multiple iterations of the magnet base at the bottom. We knew from the start that we'd have to optimize the number of magnets. On one hand, if we did not have enough magnets, Draggy would just fall off. On the other hand, if we used too many

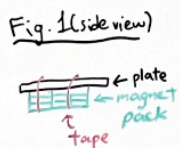
magnets, Draggy would either become too heavy, or it would prevent the wheels from spinning so much that it wouldn't be able to move.

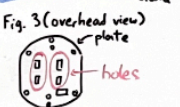
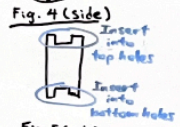
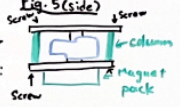
Draggy Installation/User Manual:

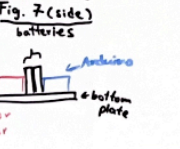
Draggy Installation Manual

The installation of Draggy requires two parts: constructing the chassis and wiring the electronics.

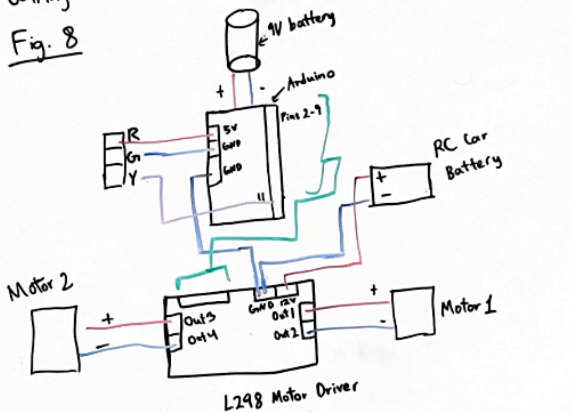
a) Installing the chassis:

- ① Attach magnet pack to the bottom plate with tape (Do not cover the 4 middle holes):

- ② Screw stands onto the motors:

- ③ Place motor stands into the aforementioned middle 4 holes:

- ④ Add another plate (top plate) onto the motor stands. Make sure the stands go into the same holes as aforementioned:

- ⑤ Place 2 columns in between the plates. Screw plates together by screwing columns into the plates:

- ⑥ Attach wheels to motor hubs. Attach inside-out rubber caps to the wheels.

- ⑦ Attach motor driver, 9V battery, Arduino, & RC Car battery using rubber bands. Attach to top plate.


b) Wiring the electronics:
Fig. 8



L298 Motor Driver

Table 1

Arduino Pin	L298 Motor Driver
2	In 1
3	In 2
4	In 3
5	In 4
6	ENA
9	ENB

Table 2

L298 Motor Driver	Other
12V	RC +
GND	RC - / GND
In 1	Pin 2
In 2	Pin 3
In 3	Pin 4
In 4	Pin 5
ENA	Pin 6
ENB	Pin 9
Out 1	Motor 1 +
Out 2	Motor 1 -
Out 3	Motor 2 +
Out 4	Motor 2 -

⑧ Wire according to Fig. 8, Table 1, and Table 2

Draggy User Manual

This manual will cover turning on/off Draggy and how to control its movement.

Turning on/off:

- ① To turn on/off Draggy, just plug in/unplug the leads from the RC car battery and plug in/unplug the Arduino - 9V battery jack.
- ② You can charge the RC car battery using the included USB charger.

↳ Note: DO NOT try charging the battery with your computer!! Use a power outlet.

↳ Note: The USB charger will glow red when charging and green when it's done.

Controlling Movement

Figure 1

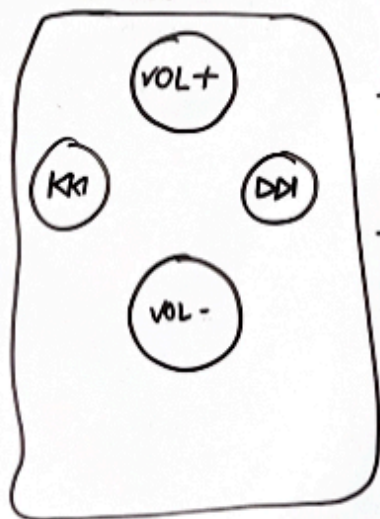


Table 1

Button	Action
VOL +	Forward
KK	Left
DD	Right
VOL -	Backward

- ③ Press the corresponding button as the desired action while pointing at the IR Receiver to both start and stop Praggy.

Data Analysis:

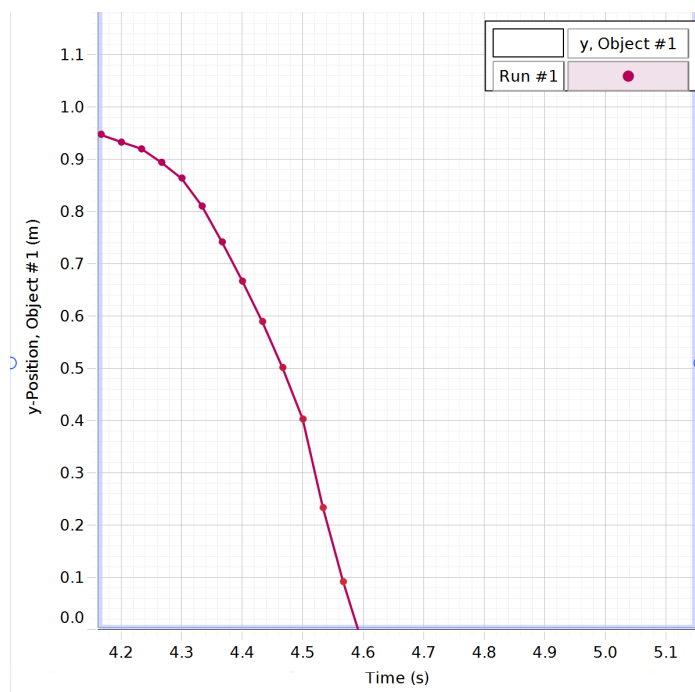
Kinematics Equations

Claim: We will prove the equation $\Delta x = \frac{1}{2} at^2$ by solving for the gravitational constant on Earth.

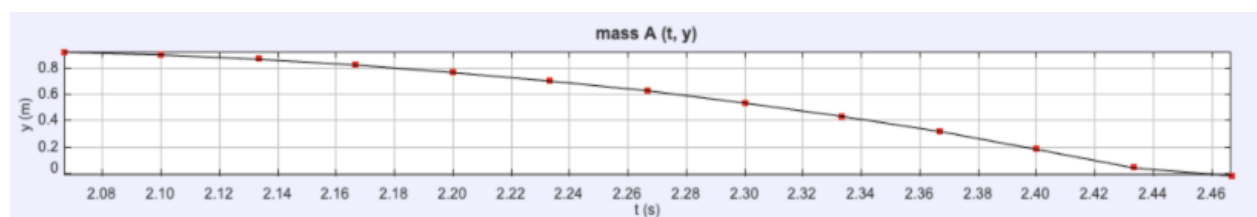
We predict that the constant will be 9.807.

Evidence: We dropped a rubber RC car wheel and tracked its position relative to time.

Trial 1



Trial 2



Trial 1		Trial 2	
	m		mass_A
4.167	0.95	t (s)	y (m)
4.2	0.93	0.00	0.92
4.233	0.92	0.03	0.91
4.267	0.89	0.07	0.88
4.3	0.86	0.10	0.85
4.333	0.81	0.13	0.80
4.367	0.74	0.17	0.74
4.4	0.67	0.20	0.68
4.433	0.59	0.23	0.60
4.467	0.5	0.27	0.50
4.5	0.4	0.30	0.40
4.533	0.23	0.33	0.28
4.567	0.09	0.37	0.16
		0.40	0.01
		0.43	0.00
t=	delta x =	Delta t:	Delta x:
0.4	0.86	0.43	0.92
0.217156	2.06	0.1849	1.84
x = 1/2 a t^2		x = 1/2 a t^2	
9.486267936		9.951325041	
	Avg g		
	9.718796488		

Reasoning: We plugged the data that we collected into the equation $\Delta x = \frac{1}{2}at^2$ and solved for the **a** value. We then averaged out the 2 values that we got, 9.486 and 9.951, and that averages out to an **a** value of 9.719, very close to the accepted value of 9.807.

Conclusion: Our expected value was 9.807, and our experimental value was 9.719. This is a difference of 0.088, which translates to a percent error of less than 1% (0.897% to be exact).

Such a small difference can be attributed to human error, OR, since one of the assumptions we made at the beginning of this experiment was negligible air resistance, the error could also come from the drag force. Since we are able to solve for the gravitational constant with the equation

$\Delta x = \frac{1}{2}at^2$, the kinematic equation holds true.

Three Laws of Motion (First and Second), Terminal Velocity, and Drag Force

Claim: There exists a terminal velocity at which the gravitational force and a drag force lead to a zero net force.

Evidence:

Trial 1				
		mass_A		
t (s)	Change in t	y (m)	Change in y	V (Change y/Change t)
0.00	0.02	0.89	-0.10	-5.588235294
0.02	0.02	0.80	-0.07	-4.125
0.03	0.02	0.73	-0.07	-4.058823529
0.05	0.02	0.66	-0.07	-4.352941176
0.07	0.02	0.59	-0.08	-5
0.08	0.02	0.51	-0.08	-4.470588235
0.10	0.02	0.43	-0.08	-4.529411765
0.12	0.02	0.35	-0.08	-5.25
0.13	0.02	0.27	-0.08	-4.941176471
0.15	0.02	0.19	-0.09	-5.252941176
0.17		0.10	Avg Velocity	-4.756911765

Trial 2				
		mass_A		
t (s)		y (m)		
0.00	0.02	0.87	-0.06	-3.764705882
0.02	0.02	0.81	-0.07	-4.625
0.03	0.02	0.74	-0.07	-4.352941176
0.05	0.02	0.66	-0.08	-4.647058824
0.07	0.02	0.58	-0.08	-4.875
0.08	0.02	0.50	-0.09	-5.117647059
0.10	0.02	0.42	-0.08	-4.823529412

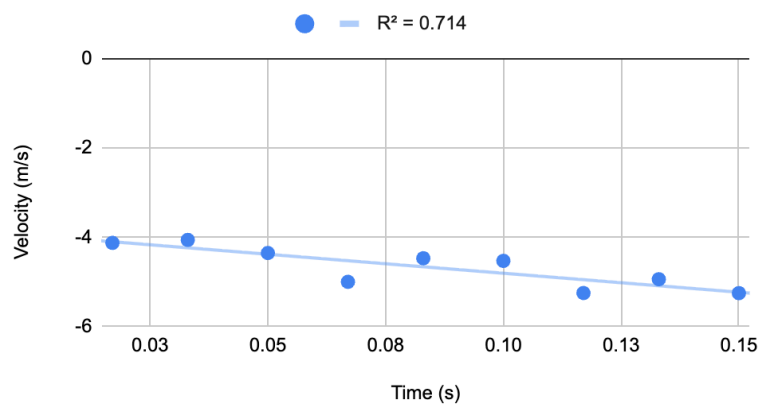
0.12	0.02	0.34	-0.09	-5.625
0.13	0.02	0.25	-0.09	-5.176470588
0.15	0.02	0.16	-0.10	-5.786470588
0.17		0.06	Avg Velocity	-4.879382353

Trial 3				
		mass_A		
t (s)		y (m)		
0.00	0.02	0.87	-0.07	-4.176470588
0.02	0.02	0.80	-0.07	-4.5
0.03	0.02	0.73	-0.08	-4.411764706
0.05	0.02	0.65	-0.08	-4.705882353
0.07	0.02	0.57	-0.08	-5
0.08	0.02	0.49	-0.08	-4.705882353
0.10	0.02	0.41	-0.09	-5.058823529
0.12	0.02	0.33	-0.09	-5.625
0.13	0.02	0.24	-0.09	-5.352941176
0.15	0.02	0.15	-0.09	-5.098823529
0.17		0.06	Avg Velocity	-4.863558824

Average velocity across three trials: $-4.833284314 \frac{m}{s}$

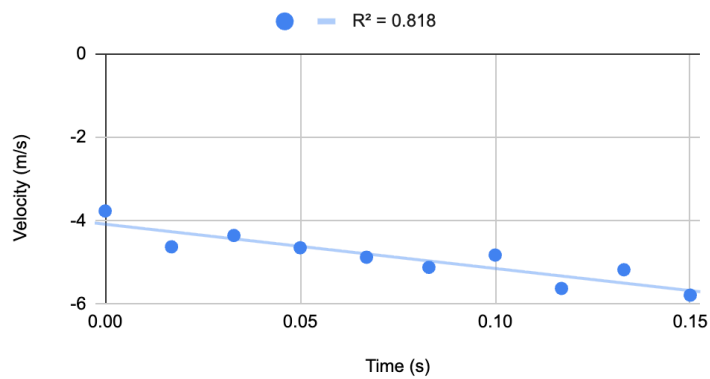
Trial 1:

Time (s) vs. Velocity (m/s)



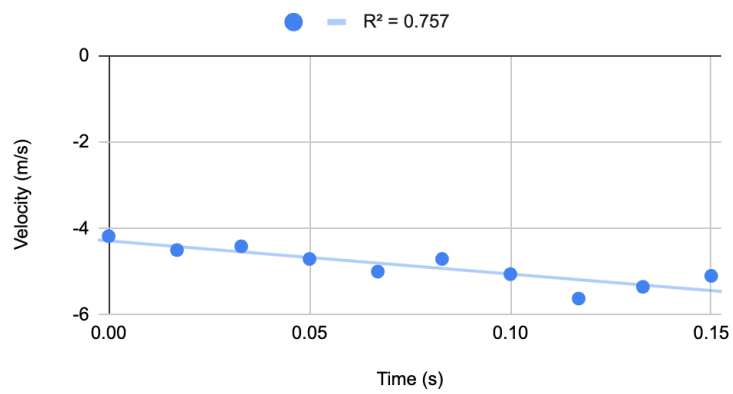
Trial 2:

Time (s) vs. Velocity (m/s)



Trial 3:

Time (s) vs. Velocity (m/s)



$$0 = F_g - F_d$$

$$F_g = F_d$$

$$mg = bv$$

$$(25.37 \text{ g}) \left(-9.81 \frac{\text{m}}{\text{s}^2}\right) = b \left(-4.833 \frac{\text{m}}{\text{s}}\right)$$

$$B = 51.4928 \frac{\text{g}}{\text{s}}$$

Reasoning:

The velocity of the wheel as it dropped remained around $4.833 \frac{m}{s}$. Assuming this was constant, this is the terminal velocity. This meant there was no acceleration, and according to the first law of motion, there is no net force on the wheel. No net force leads to the implication that there is a drag force that is acting upon the wheel, and it is equal to the force of gravity at this instance.

From the second law of motion, $F = ma$, we can solve for the gravitational force. Assuming it is a linear drag force with the formula Bv , we can solve for the drag constant B with the equation $0 = mg - bv$, or $mg = bv$.

Conclusion: With the velocity constant, it is the terminal velocity, and provides the implication of the drag force. Assuming a linear drag force, we can solve for the drag constant B with the knowledge of the mass of the wheel, the accepted gravitational constant g , and the terminal velocity of the wheel. This supports our claim that there is a terminal velocity where a drag force acts equal and opposite to the gravitational force.

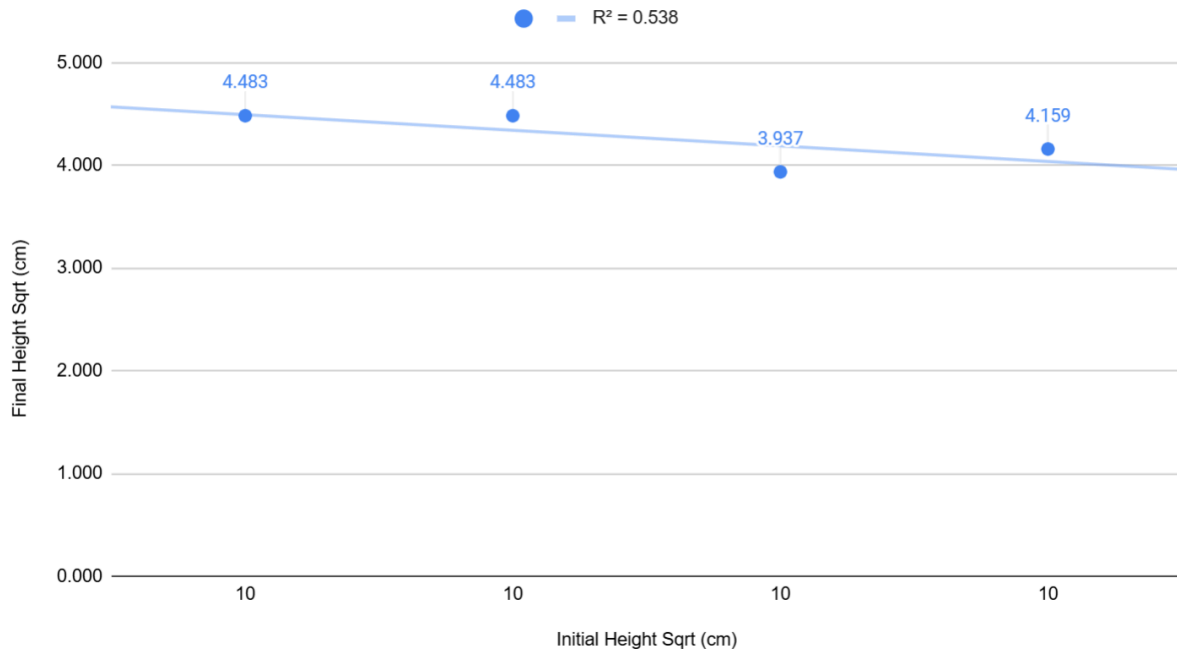
Coefficient of Restitution and Newton's Third Law of Motion

Claim: The collision between the wheel and the floor is not elastic, there is a non-trivial coefficient of restitution.

Evidence:

Trials	Initial Height cm	I Height sqrt cm	Final Height cm	F Height sqrt cm	Coeff of Rest
1	100	10	20.1	4.483	0.4483302354
2	100	10	20.1	4.483	0.4483302354
3	100	10	15.5	3.937	0.3937003937
4	100	10	17.3	4.159	0.4159326869
Average Coeff	0.4265733879				

Initial Height Sqrt (cm) v Final Height Sqrt (cm)



Computational: $e^2 K_i = K_f$

Conceptual: Momentum is conserved as long as there is no external force. If the momentum is conserved, then that means the final height of the wheel should be the same as the initial.

Reasoning:

After measuring the balls' heights before and after the bounce, we found that the height after the bounce was much lower than before. After deriving the equation for the coefficient of restitution

in terms of heights from the definition $e^2 K_i = K_f$: $e^2 = \frac{K_f}{K_i}$, where the kinetic energy before and

after are the same as the Mechanical Energy. Then, Gravitational Potential Energy is the

Mechanical Energy at the top of the trajectories when Kinetic Energy is zero. Hence $e^2 = \frac{mgh_f}{mgh_i}$

$\rightarrow e^2 = \frac{h_f}{h_i} \rightarrow e = \sqrt{\frac{h_f}{h_i}}$. The coefficient of restitution for the balls on both surfaces was less

than 1 (0.88, 0.73, 0.71, 0.72), which means that the kinetic energy is not conserved through this inelastic collision.

As we can see in this experiment, we are using Newton's third law of motion as our basis for this experiment. However, because the wheel doesn't bounce all the way back up to its original position, as it would have if the coefficient of restitution was 1 and as stated in Newton's third law, that every action has an equal and opposite reaction, we can use that information to calculate the coefficient of restitution based on the initial and final heights.

Conclusion: w

Moment Of Inertia/Center of Mass/ Conservation of Energy with Turning

Points

Claim: A physical pendulum's measured period can be used to determine its moment of inertia about the center of mass, which is approximately $1.96 \cdot 10^{-5} \text{ kg} \cdot \text{m}^2$.

Evidence:

Seconds	Oscillations	Period	I_{edge}	Average I_{edge}	I_{com}
5.35	11	0.4863636364	0.0000298	0.0000298	0.0000196
5.36	11	0.4872727273	0.0000299		
5.24	11	0.4763636364	0.0000286		
5.24	11	0.4763636364	0.0000286		
7.59	15	0.506	0.0000322		

Computational:
$$I_{com} = \frac{T^2 mgd_{com}}{4\pi^2} - md_{com}^2$$

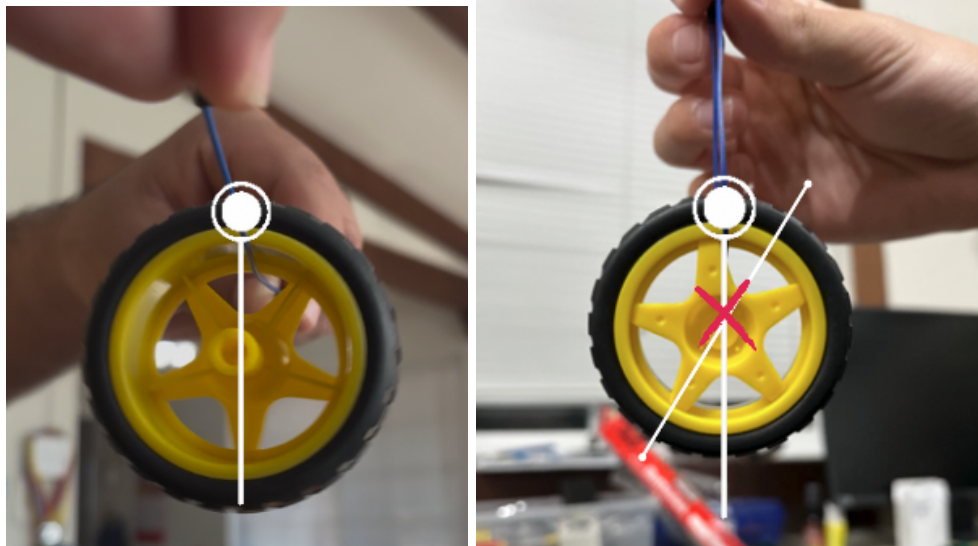
Conceptual: Torque can be written in terms of angular acceleration and $F \cdot d$. Equating these two forms, we can get a second-order differential equation from which we can glean the angular

velocity. From the relation between period and angular velocity, we can derive the moment of

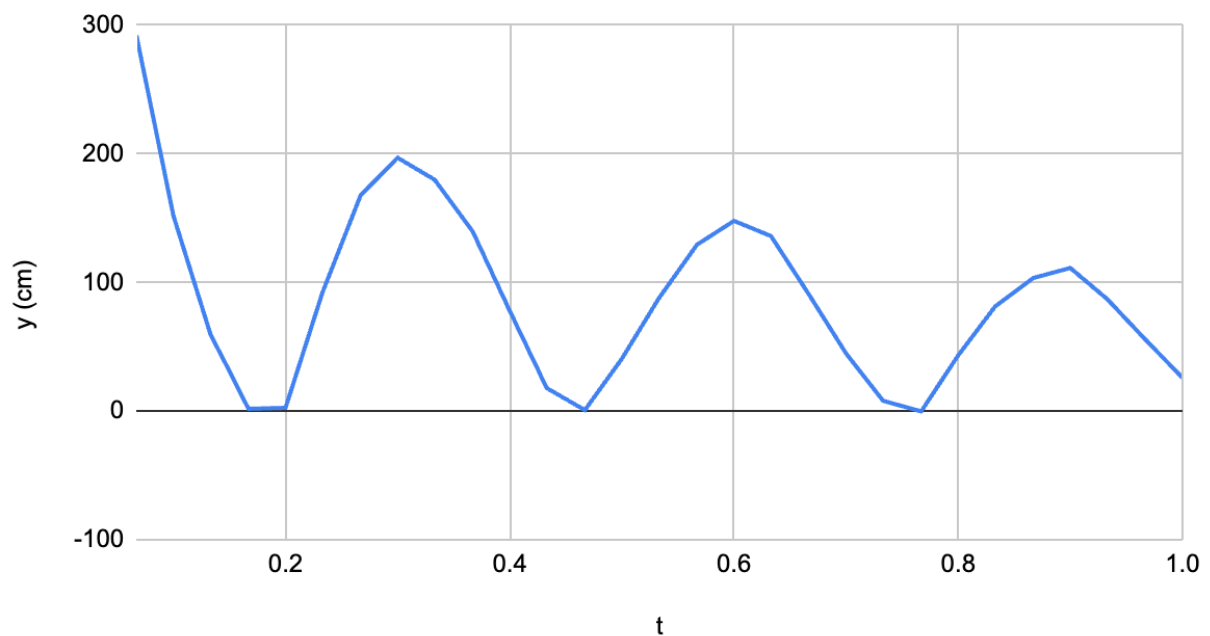
inertia in terms of period. Then, using the parallel axis theorem, $I_{edge} = I_{com} + md_{com}^2$,

remove md_{com}^2 from both sides to get the equation above.

Visual (Hanging method for finding Center of Mass):



y (cm) vs. t (s)



Reasoning:

The moment of inertia we calculated comes directly from how a physical pendulum behaves. The period of a physical pendulum depends both on its mass distribution and distance of the pivot from the center of mass. Across the five trials, period measurements stayed within a small range of 0.48-0.51 seconds. The resulting values of I_{edge} , or the moment of inertia about a pivot 2 cm from the center of mass, stayed rather consistent, as well as with only one minor outlier. Applying the parallel axis theorem, we get values for I_{com} consistent for a semi-solid disk of radius 1 inch ($1.96 \cdot 10^{-5} \text{ kg}\cdot\text{m}^2$).

The graph above shows the y-coordinate of a point on the physical pendulum. Since $GPE = mgy$, scaling the graph by a constant amount yields a Potential vs. time graph. This graph, through turning points, can be analyzed to find the stable and unstable equilibria of this physical pendulum system. Since every odd extremum has a second derivative that is positive ($P''(t_{crit, odd}) > 0$), we can conclude that at times $t_{crit, odd}$, there's a stable equilibrium. Similarly, since every even extremum has a second derivative that is negative ($P''(t_{crit, even}) < 0$), we can conclude that at times $t_{crit, even}$, there's an unstable equilibrium.

Conclusion: w**Rotational kinematics, dynamics, and energies**

Claim: We can calculate the velocity of the wheel given its angular velocity and radius.

Evidence:

Height: 27 cm = 0.27 m

Radius of wheel: 3.25 cm = 0.00325 m

Assumption for energy calculation: no friction

Initial energy calculation = $mgh = (25.37g)(9.81\frac{m}{s^2})(0.27m) = 67.197519 \text{ N}$

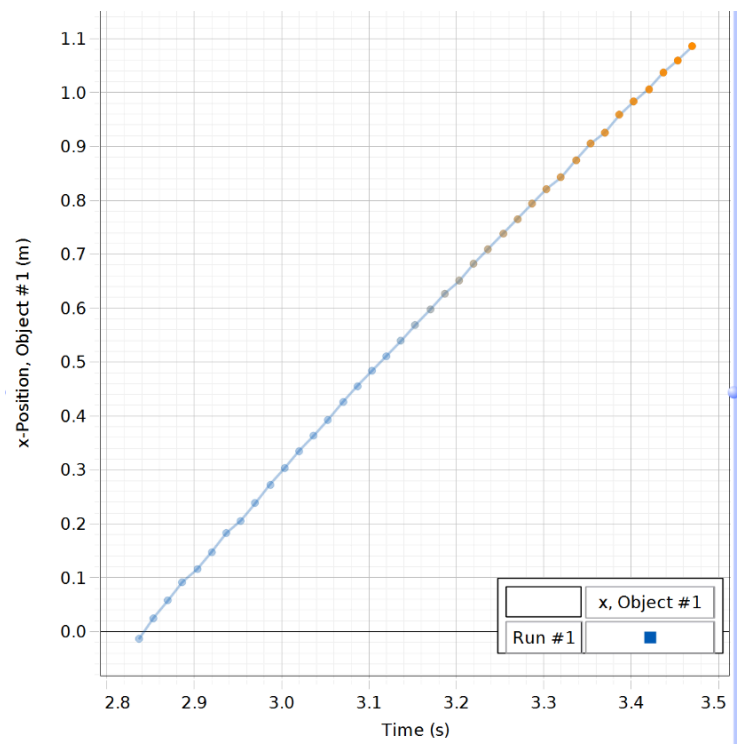
Initial Velocity @ bottom of ramp = $E = \frac{1}{2}mv^2 \rightarrow v = \sqrt{\frac{2E}{m}} = 2.3016 \frac{m}{s}$

Experimental:

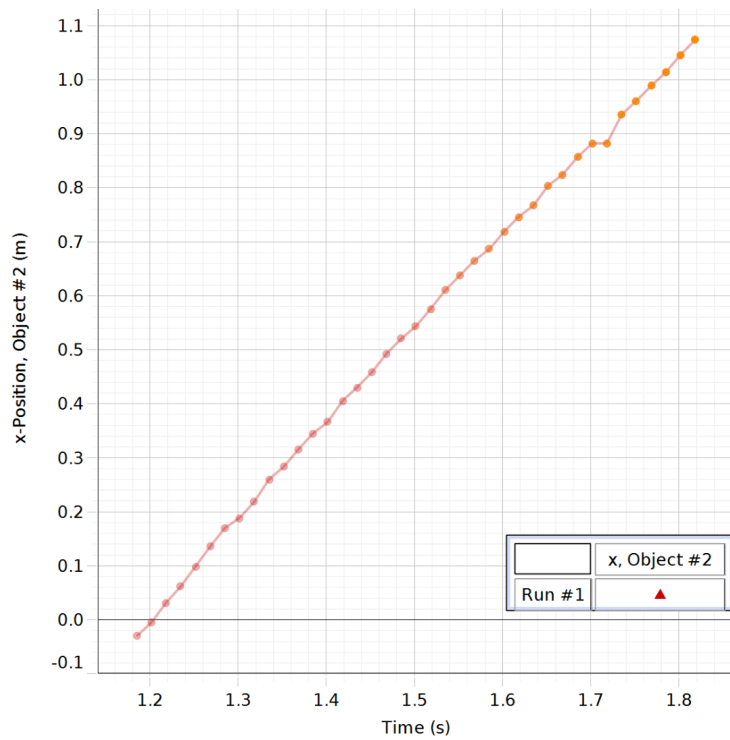
Time (s) 1	X pos	V 1	Time (s) 2	X pos	V 2	Time (s) 3	X pos	V 3
2.837	-0.01	1.7377567	1.185	-0.03	1.737756714	1.368	0.03	1.683333
2.853	0.03		1.202	0		1.385	0.06	
2.87	0.06		1.218	0.03		1.402	0.09	
2.887	0.09		1.235	0.06		1.418	0.12	
2.903	0.12		1.252	0.1		1.435	0.15	
2.92	0.15		1.268	0.14		1.452	0.19	
2.937	0.18		1.285	0.17		1.468	0.21	
2.953	0.21		1.302	0.19		1.485	0.24	
2.97	0.24		1.318	0.22		1.502	0.27	
2.987	0.27		1.335	0.26		1.518	0.3	
3.003	0.3		1.352	0.28		1.535	0.33	
3.02	0.34		1.368	0.32		1.552	0.36	
3.037	0.36		1.385	0.35		1.568	0.39	
3.053	0.39		1.402	0.37		1.585	0.42	
3.07	0.43		1.418	0.41		1.602	0.44	
3.087	0.46		1.435	0.43		1.618	0.47	
3.103	0.48		1.452	0.46		1.635	0.5	
3.12	0.51		1.468	0.49		1.652	0.53	
3.137	0.54		1.485	0.52		1.668	0.55	
3.153	0.57		1.502	0.54		1.685	0.58	
3.17	0.6		1.518	0.58		1.702	0.61	
3.187	0.63		1.535	0.61		1.718	0.63	
3.203	0.65		1.552	0.64		1.735	0.66	
3.22	0.68		1.568	0.67		1.752	0.69	
3.237	0.71		1.585	0.69		1.768	0.71	
3.253	0.74		1.602	0.72		1.785	0.74	
3.27	0.77		1.618	0.75		1.802	0.76	

3.287	0.79		1.635	0.77		1.818	0.8	
3.303	0.82		1.652	0.8		1.835	0.82	
3.32	0.84		1.668	0.82		1.852	0.85	
3.337	0.87		1.685	0.86		1.868	0.87	
3.353	0.9		1.702	0.88		1.885	0.9	
3.37	0.93		1.718	0.88		1.902	0.93	
3.387	0.96		1.735	0.94		1.918	0.95	
3.403	0.98		1.752	0.96		1.935	0.98	
3.42	1.01		1.768	0.99		1.952	1.01	
3.437	1.04		1.785	1.01		1.968	1.04	
3.453	1.06		1.802	1.05				
3.47	1.09		1.818	1.07				

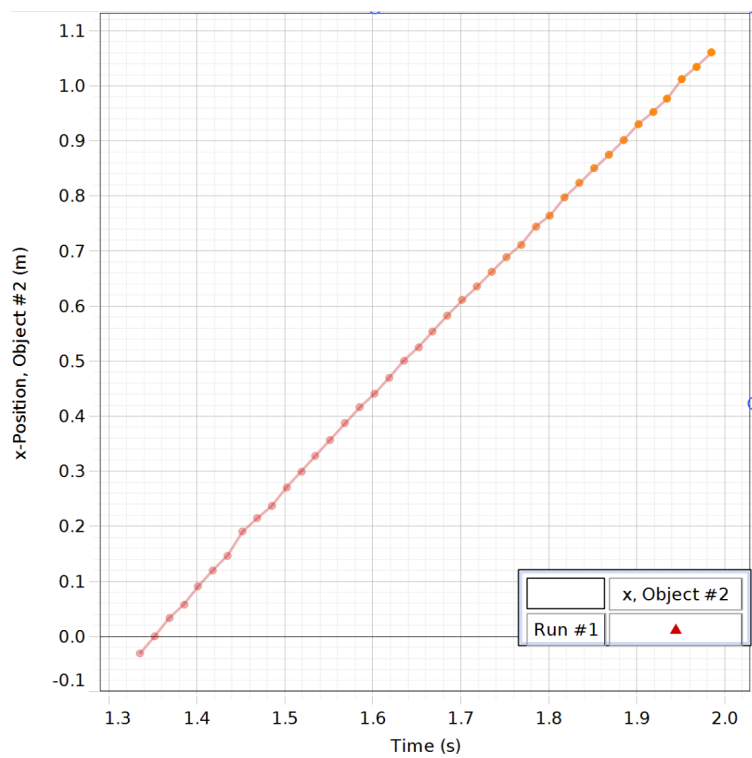
Trial 1:



Trial 2:



Trial 3:



Experimental Avg Velocity: $1.719615587 \frac{m}{s}$

Time for one rotation: $0.52 \text{ sec (slow-mo)} / 4 \text{ (time factor)} = 0.13 \text{ s}$

$$\text{Linear Calculated Velocity: } \omega r = \frac{2\pi r}{T} = \frac{2(0.0325)\pi}{0.13} = 1.5708 \frac{m}{s}$$

Reasoning:

We first calculated the velocity of the wheel if there was 0 friction to give us a floor to expect. We did this by first calculating the total potential energy the wheel had when it was at the top of the ramp, 27 cm high. Then, since there is no friction, we can assume that all of the gravitational potential energy transforms to the kinetic energy of the wheel. By doing our calculations, $mgh = \frac{1}{2}mv^2$, and then solving for v, we get $2.3016 \frac{m}{s}$.

However, we know that friction exists, and this velocity is just a floor for us to get a general gist of what our experimental velocity could be. We first recorded the time it would take for the wheel to spin one rotation after reaching the bottom of the ramp and used that to find the calculated linear velocity using the equation $\frac{2\pi r}{T}$, giving us a result of $1.5708 \frac{m}{s}$.

To get the experimental velocity, we tracked the position of the wheel on the x-axis relative to time to give us a position v time graph. I then took the positions of the first and last data points to get the average velocity of each trial, then took the averages of all the trials to get the average experimental velocity, which was $1.720 \frac{m}{s}$.

Harmonic motion with damping

Claim: The rotation of the wheel follows Simple Harmonic Motion, and the force of friction causes underdamping.

Evidence:

$$y(t) = A \cos(\omega t)$$

Radius of wheel: 4 centimeters

Period: 0.41 seconds

@ $t = 0$, $y(t) = 4$ cm

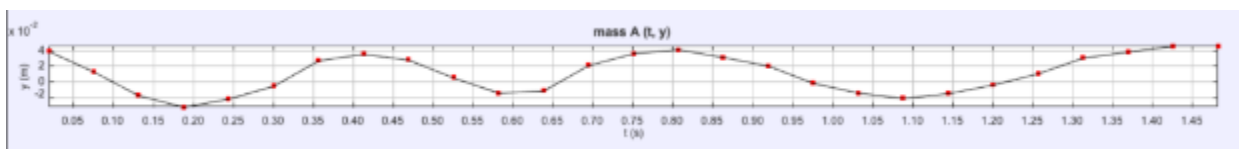
$$A = 4, \omega = \frac{2\pi}{0.41} = 15.3248$$

$$\Rightarrow y(t) = 4 \cos(15.3248t)$$

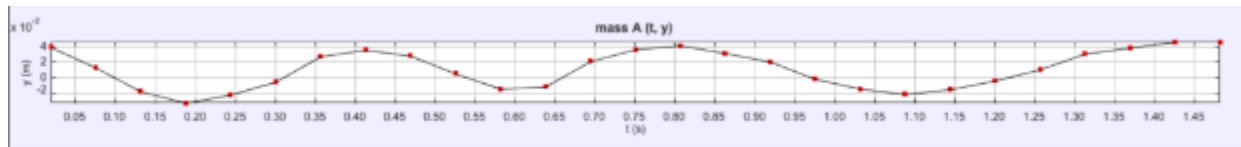
Time 1 (s)	y position(m)	Time 2 (s)	y position(m)	Time 3 (s)	y position(m)
0.02	0.04	0.00	0.03	0.00	0.04
0.08	0.01	0.05	0.01	0.05	0.01
0.13	-0.02	0.11	-0.02	0.10	-0.03
0.19	-0.03	0.16	-0.04	0.15	-0.04
0.24	-0.02	0.22	-0.04	0.20	-0.04
0.30	0.00	0.27	-0.02	0.25	-0.01
0.36	0.03	0.32	0.01	0.31	0.02
0.41	0.04	0.38	0.03	0.36	0.03
0.47	0.03	0.43	0.03	0.41	0.02
0.53	0.01	0.48	0.01	0.46	0.00
0.58	-0.01	0.54	-0.01	0.51	-0.03
0.64	-0.01	0.59	-0.03	0.56	-0.04
0.69	0.02	0.65	-0.04	0.61	-0.04
0.75	0.04	0.70	-0.03	0.66	-0.03
0.81	0.04	0.75	-0.01	0.71	0.00
0.86	0.03	0.81	0.02	0.76	0.01
0.92	0.02	0.86	0.03	0.81	0.03
0.98	0.00	0.91	0.04	0.86	0.03
1.03	-0.01	0.97	0.03	0.91	0.02
1.09	-0.02	1.02	0.01	0.96	0.00
1.14	-0.01	1.08	-0.01	1.02	-0.02
1.20	0.00	1.13	-0.02	1.07	-0.03
1.26	0.01	1.18	-0.03	1.12	-0.04
1.31	0.03	1.24	-0.03	1.17	-0.04

1.37	0.04	1.29	-0.02	1.22	-0.03
1.43	0.05	1.34	-0.01	1.27	-0.01
1.48	0.05	1.40	0.00	1.32	0.01
		1.45		1.37	0.02
		1.51		1.42	0.03
		1.56		1.47	0.03
		1.61		1.52	0.02
		1.67		1.57	0.01
		1.72		1.62	0.00
		1.77		1.68	-0.01
		1.83		1.73	-0.03
		1.88		1.78	-0.03
		1.94		1.83	-0.04
		1.99		1.88	-0.04
		2.04		1.93	-0.03
		2.10		1.98	-0.03
		2.15		2.03	-0.02
		2.20		2.08	-0.01
		2.26		2.13	0.00
		2.31		2.18	0.01
		2.37		2.23	0.02
		2.42		2.28	0.02
		2.47		2.34	0.03
		2.53		2.39	0.03
		2.58		2.44	0.03
		2.63		2.49	0.03
		2.69		2.54	0.03
		2.74		2.59	0.03

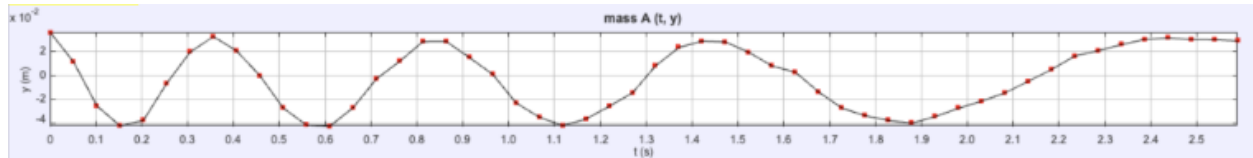
Trial 1



Trial 2



Trial 3



Reasoning:

The wheel has a radius of approximately 4 centimeters, and therefore, the amplitude of the sinusoidal curve created is 4 centimeters because that is the furthest distance it will be from the horizontal baseline going through the center. Since we began tracking from the top of the wheel, the sinusoidal curve will be graphed by a cosine wave. The period of the wave, or the time it takes for the point to make a full circle, is 0.41 seconds. This can be used to calculate because the angular velocity ω is equal to $\frac{2\pi}{T}$. This leads to the equation $y(t) = 4\cos(15.3248t)$, which models the harmonic motion of the point on the wheel.

Conclusion: Then, the graphs of the trials model a graph of underdamping. As time goes on, the wheel begins to slow down, and each cycle's period becomes longer. The graph finally begins petering out to a flat line as the point on the wheel stays constant. This supports the claim that a single point on the wheel follows a simple harmonic motion that is affected by the underdamping of friction.

Sources of Error:

Error can be derived from an imperfect wheel that has deformation and friction. There are some human errors in tracking data points, such as shakiness in the video and skewed perspectives in the videos. The tracked videos may also error in keeping data points constant.

Final Assumptions & Limitations:

Finally, we assume the water bottle boots on the wheels will stay lined on the wheel. With the IR remote/receiver, we must assume that there will be no interference from other signals during our demonstration. We must also assume that the remote signal will be oriented and aligned well enough for the receiver to get the signal to start and stop.

The sponge used to erase the marker is a porous material that is very malleable. However, this flexibility of the sponge presents a difficulty in deciding and cutting the thickness of the sponge needed. On one hand, a thin sponge will not erase the marker as well, but on the other hand, a thick sponge will apply pressure and push Draggy off the whiteboard. We are limited by the tools to cut the sponge with precision, so we have cut it thin to maintain Draggy on the whiteboard at the sacrifice of eraser effectiveness.

Skills We Learned/Improved:

- **Computer-Aided Design (CAD):** Using OnShape, we designed several possible versions of Draggy. While we originally planned to print a chassis using 3D printing, we eventually settled on the quicker and stronger acrylic laser-cutting. We learned to use a software program called Glowforge, and learned how to convert images to and use bitmaps.

- **Laser-cutting:** Using Glowforge, we experimented with laser-cutting different materials, ranging from acrylic to wood to leather. We also learned how to utilize the “engrave” and “3D-engrave” features, which could allow us to improve the aesthetics of our robot.
- **Soldering:** While our original, high RPM motors came with included jumper wires, our new motors did not. We initially experimented with temporary connections, but knew that soldering would be the most reliable way to utilize the motors. We learned how to solder wires together, and even more, how to quickly and reliably, strip away insulation.
- **Coding:** We utilized Arduino’s modified C++ language in order to send signals to the motor driver, which controls the motors independently, through the Arduino. We also integrated remote control logic using the IRremote Library in the Arduino Integrated Development Environment (IDE) and IR remote/receiver pairs.
- **Reading schematics and circuit diagrams:** We used an L293D H-bridge to upload our predesignated route in Draggy. At first, we didn’t know how to connect everything or if the placement of the wires mattered on the circuit board. But with more experimentation in different positions and connections, we quickly learned that although it looks complicated, if you work on a section of the diagram one at a time, it will slowly make sense.
- **Learning to use the multimeter:** To ensure safety, as well as understand the electrical requirements of Draggy, we learned how to better utilize the multimeter. Firstly, we began using different connections to the multimeter other than the traditional probes, like alligator clips and hook attachments. Secondly, we learned how to measure current draw under load.

Bibliography:

DUAL FULL-BRIDGE DRIVER Multiwatt15. 2000.

L293, L293D QUADRUPLE HALF-H DRIVERS D Featuring Unitrode L293 and L293D

Products Now from Texas Instruments D Wide Supply-Voltage Range: 4.5 v to 36 v D

Separate Input-Logic Supply D Internal ESD Protection D Thermal Shutdown D

High-Noise-Immunity Inputs D Functionally Similar to SGS L293 and SGS L293D D

Output Current 1 a per Channel (600 MA for L293D) D Peak Output Current 2 a per

Channel (1.2 a for L293D) D Output Clamp Diodes for Inductive Transient Suppression

(L293D) Description/Ordering Information.

“Wipy: The Overly Motivated Whiteboard Cleaner.” *Instructables*,

www.instructables.com/Wipy-the-Overly-Motivated-Whiteboard-Cleaner/.